

Abstract geometric lines in black, forming various polygons and intersecting patterns, primarily located on the left side of the slide.

MATHEMATICAL ART ON THE AXIDRAW PEN PLOTTER

Christopher Hanusa

Queens College of the City University of New York

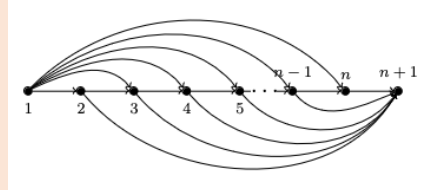
qc.edu/~chanusa

@mathzorro

ABOUT ME

Mathematician

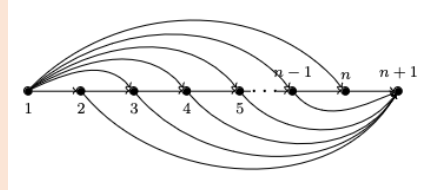
- Combinatorics
- Flow Polytopes



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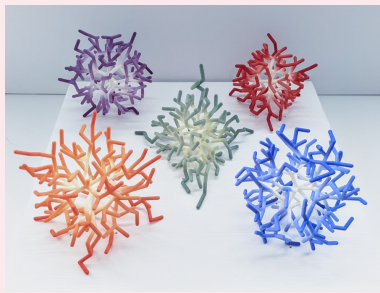
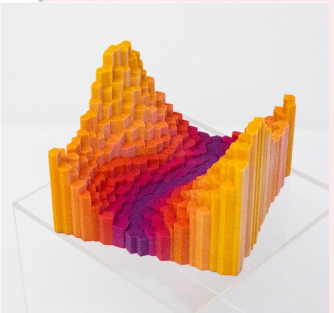
Mathematician

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Mathematical Artist

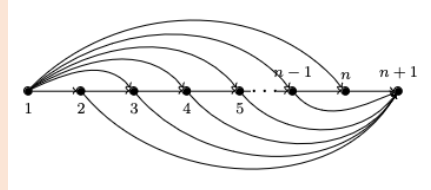
- 2D and 3D Visualizations
- Generative Art



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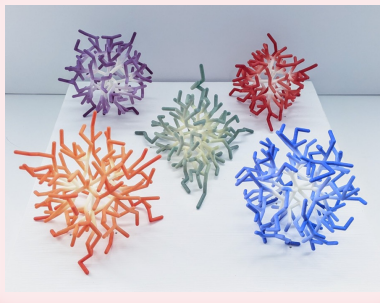
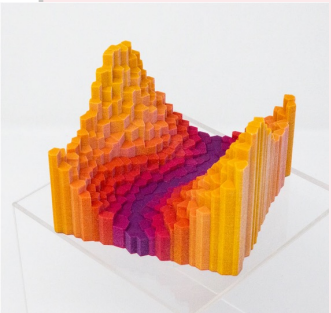
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Jewelry Designer

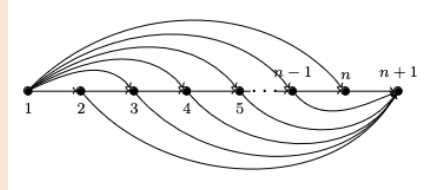
- Hanusa Design
MATHEMATICAL JEWELRY
- Museum Shops



ABOUT ME

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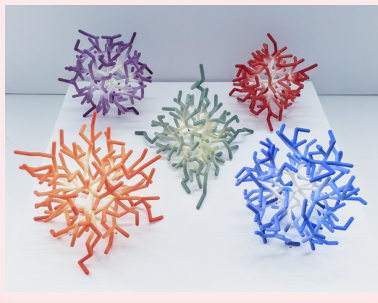
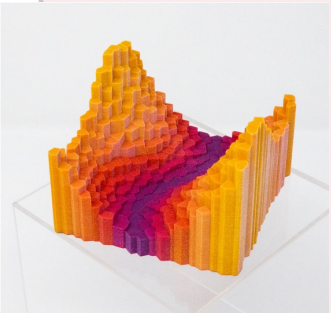


Mathematics Educator

- Course design and development
- Experiential, Interdisciplinary & Student-Centered Teaching
- Projects, Writing, Coding

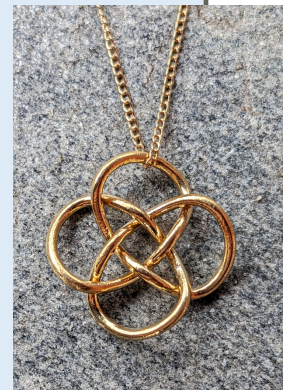
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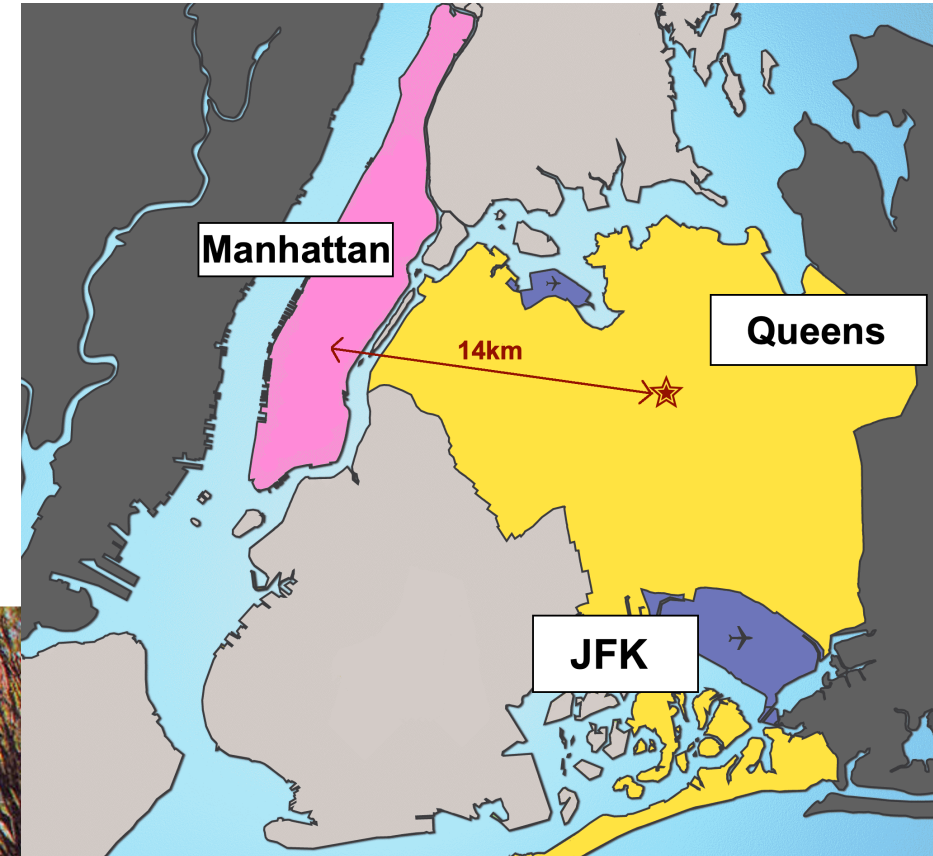
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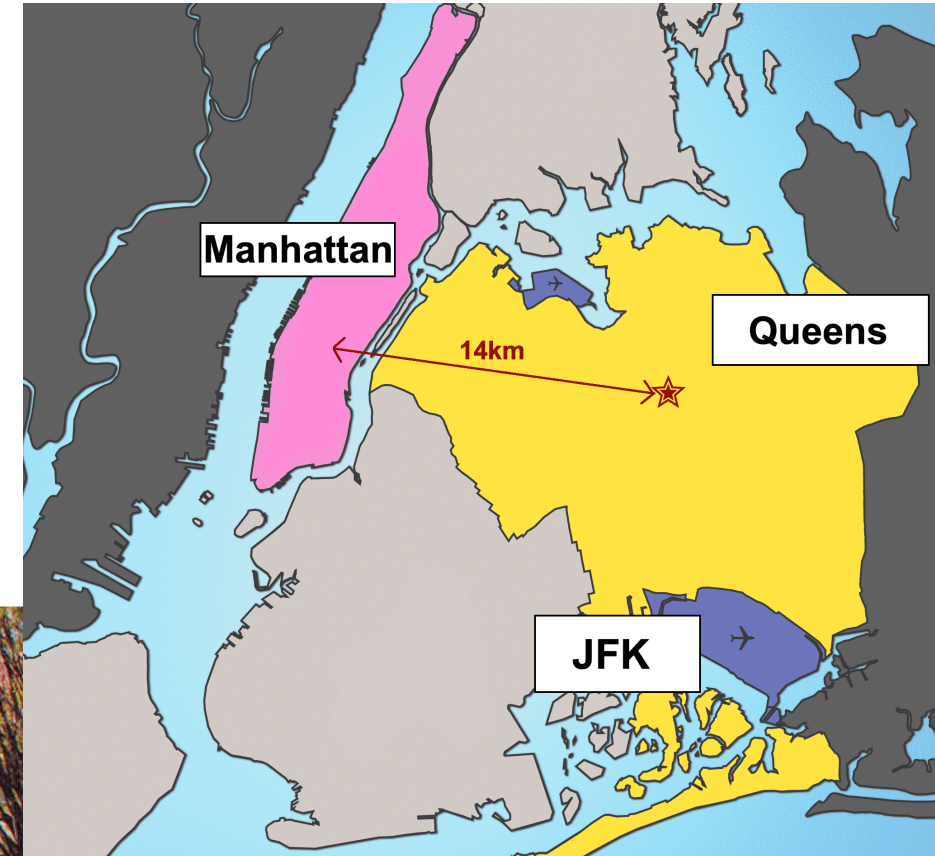
ABOUT QUEENS COLLEGE

- **Urban Commuter Campus**



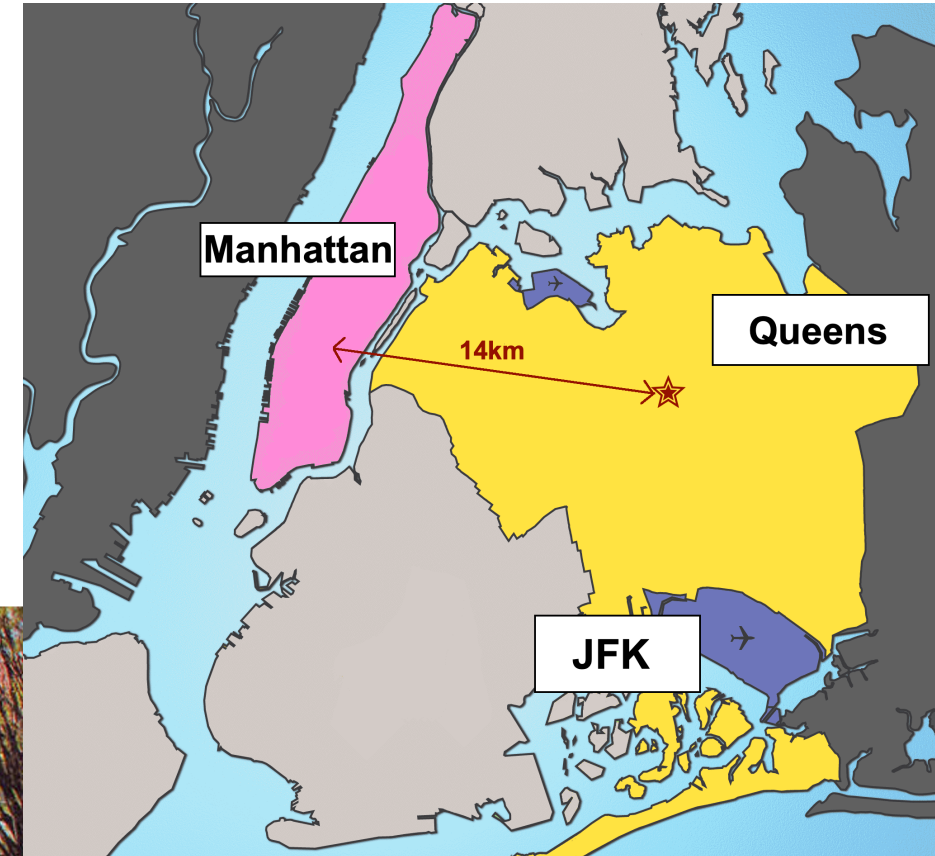
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- **Urban Commuter Campus**
- **Diverse Student Population**
 - 17,000 students
 - 100 countries
 - 35% First Generation Students



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- **Urban Commuter Campus**
- **Diverse Student Population**
 - 17,000 students
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 - 35% First Generation Students
- **Liberal Arts Education**



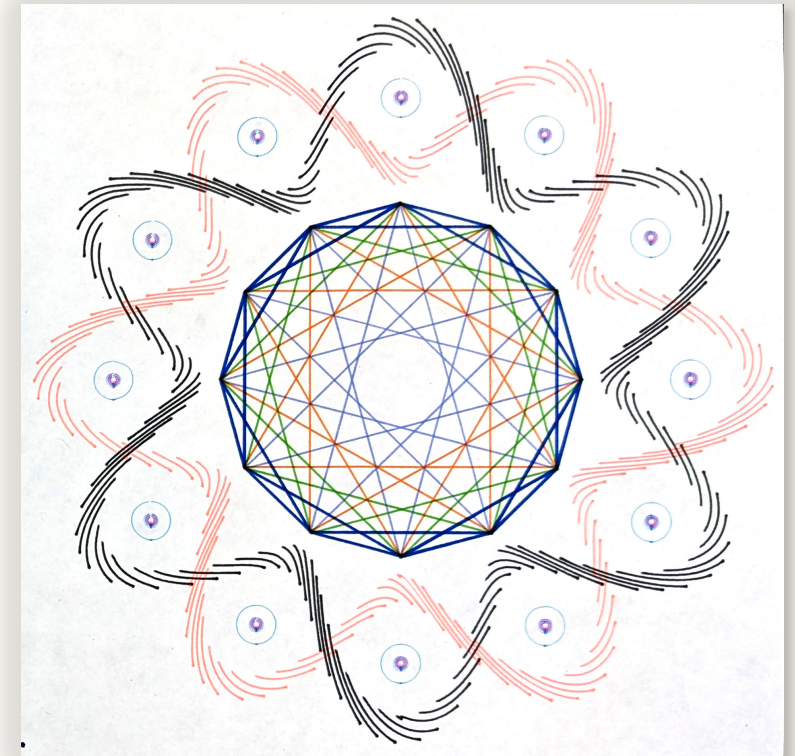
MATH 128: MATHEMATICAL DESIGN

New course for students NOT studying mathematics

Goal: Explore and appreciate the **beauty of mathematics** through a hands-on experience

Tools:

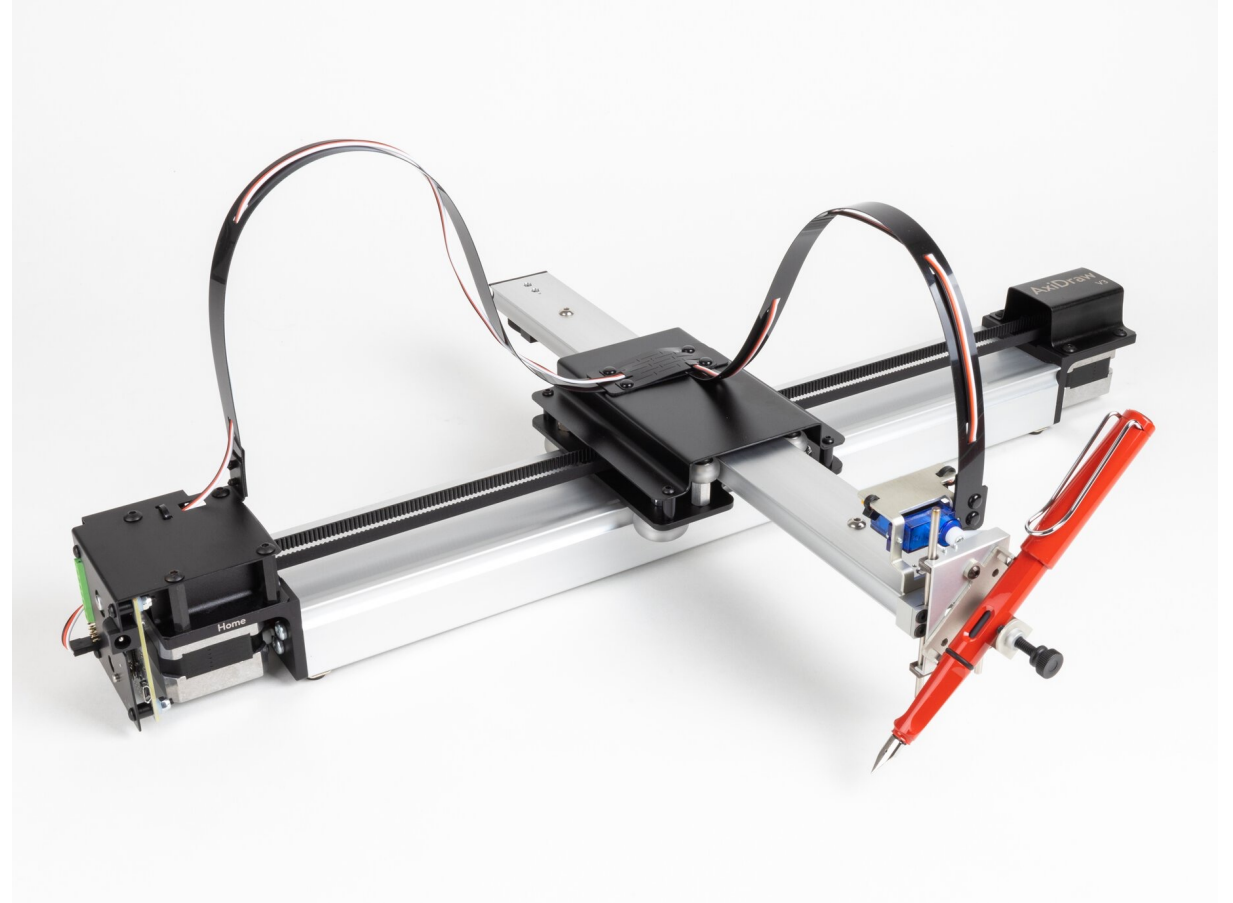
- Desmos Classroom
- AxiDraw Pen Plotter



Weize Zheng Spr. 2023

AXIDRAW PEN PLOTTER

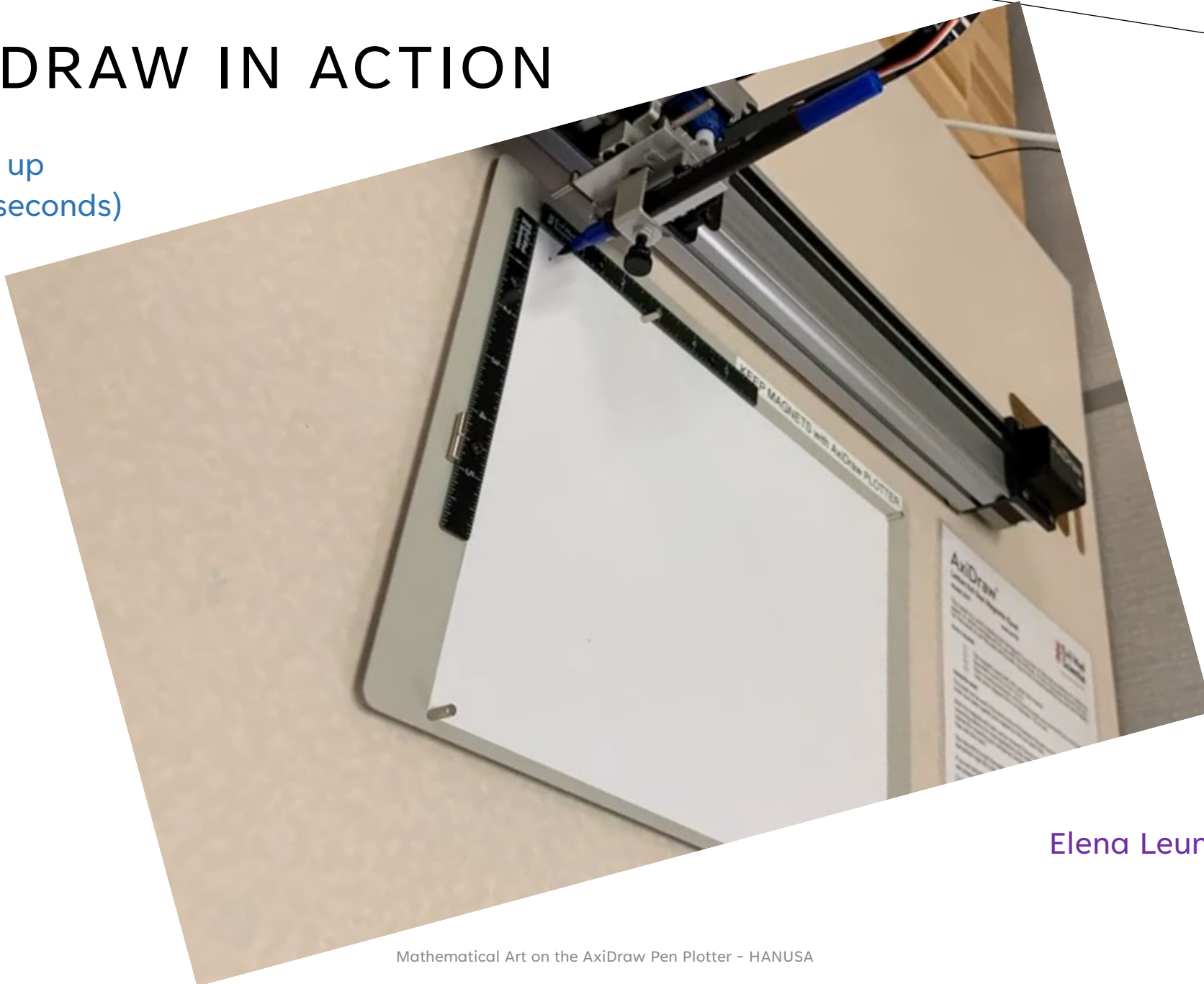
- Pen moves along x- and y-axes & (up-down)
- Choice of writing tool
- Precisely replicates digital drawings



AxiDraw v3 by Evil Mad Scientist Labs, axidraw.com

AXIDRAW IN ACTION

Video sped up
(2 minutes > 10 seconds)



Elena Leung, Spring 2023

EXPLORER MINDSET

How to get from (**afraid of math**) to (**making mathematical art**)?



EXPLORER MINDSET

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- Mathematics = “get the right answer”
- Afraid of **math** = afraid of **failing**



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- Never given the chance to **PLAY** with math.



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- Bite-sized chunks for incremental progress
- Create a **community** of learners



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Encourage **Productive Failure**

- **Explore!** Get inspired from work of others.
- What would happen if ...



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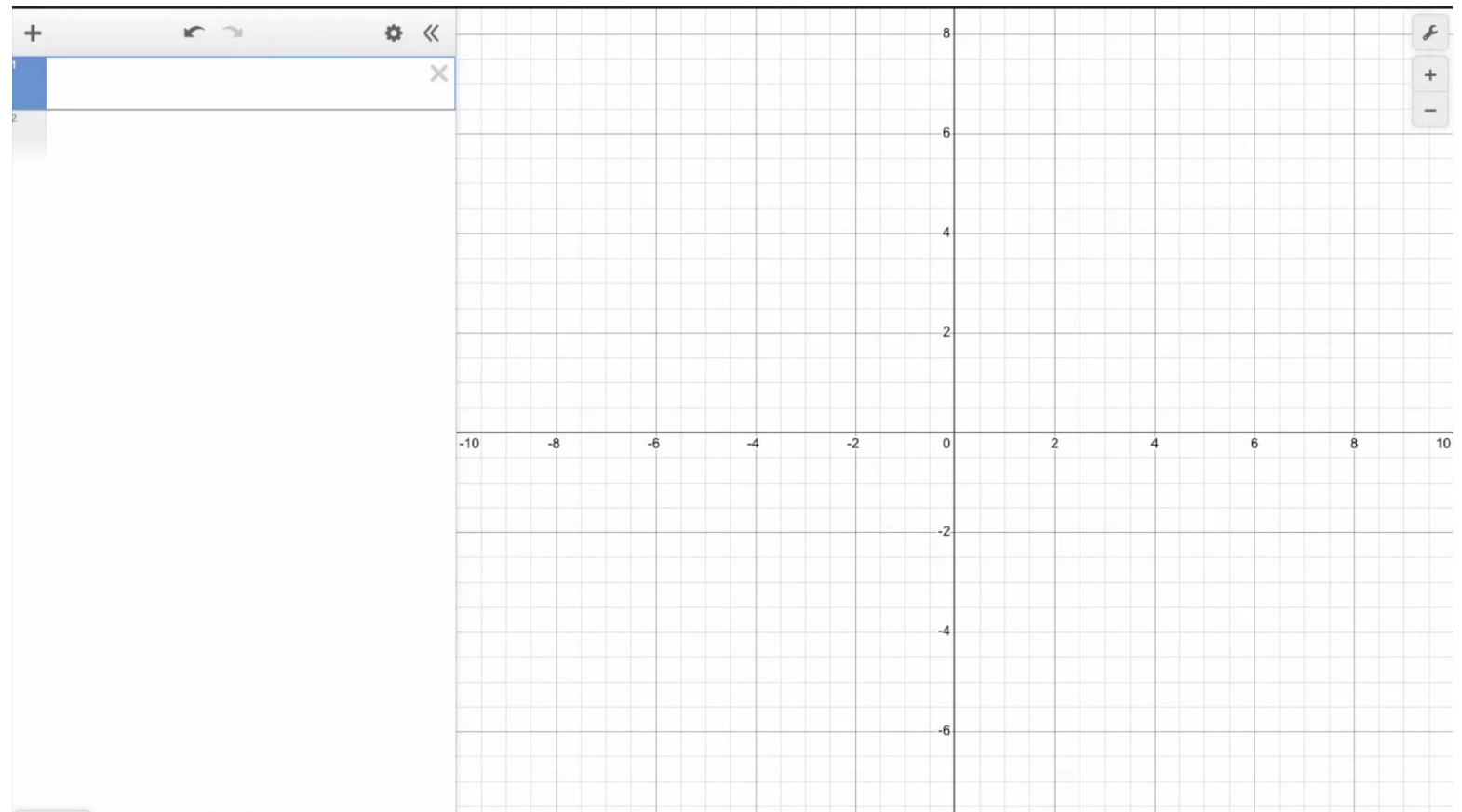
- **Explore!** Get inspired from work of others.
- What would happen if ...
- Easy in Desmos with **sliders!**



WORKFLOW: DESMOS → AXIDRAW

Build artwork in Desmos

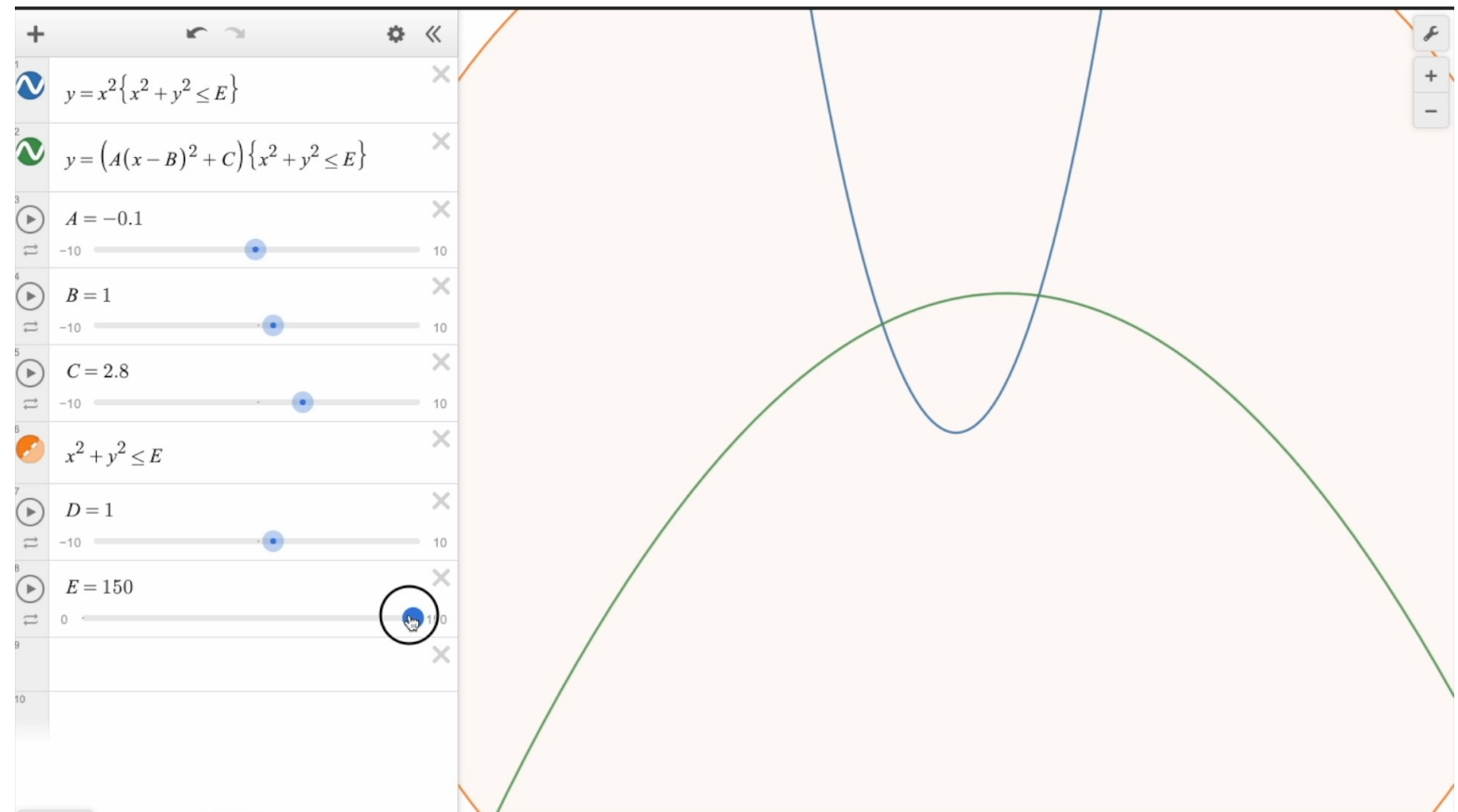
- Equations
- Transformations



WORKFLOW: DESMOS → AXIDRAW

Build artwork in Desmos

- Equations
- Transformations
- Restrictions



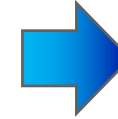
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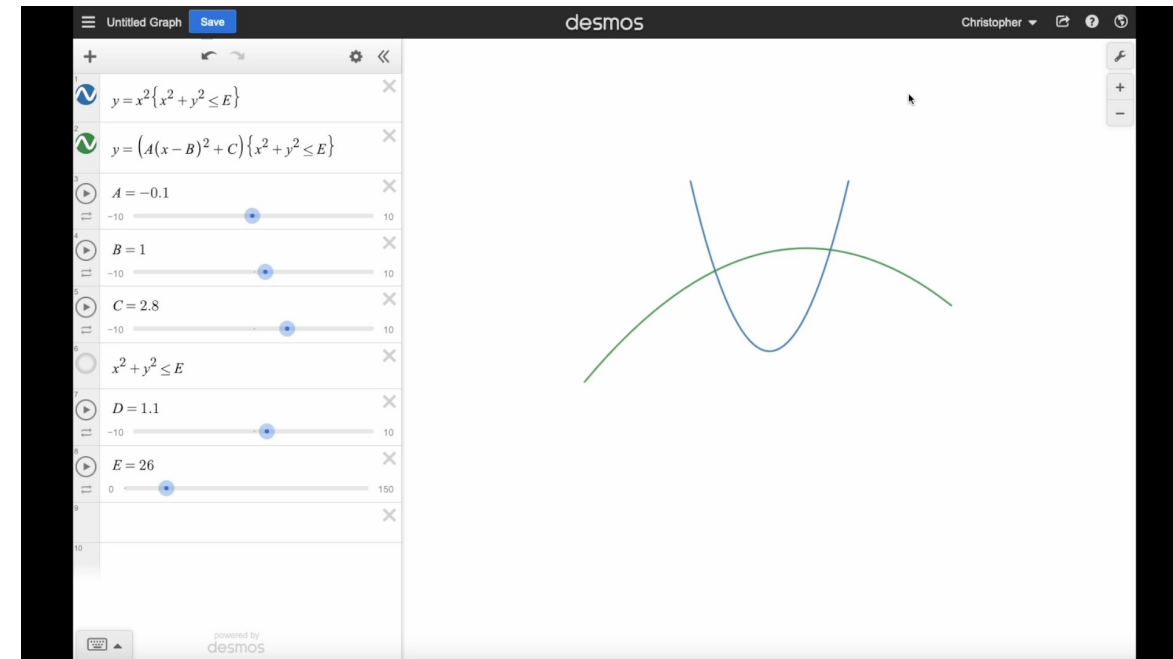
desmos



INKSCAPE
Draw Freely.

Export as SVG; Import to Inkscape

- Export from Desmos as SVG file
- Import to Inkscape and resize to canvas



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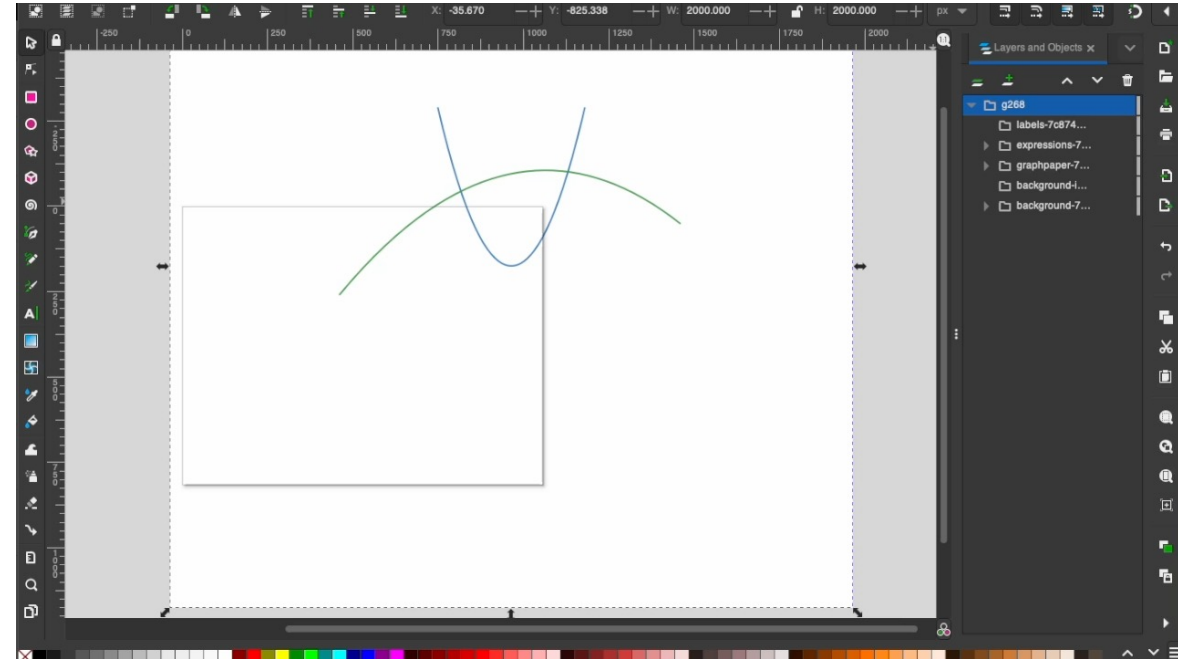
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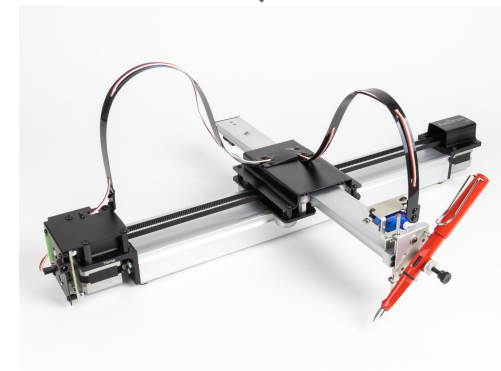
- Export from Desmos as SVG file
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Prepare AxiDraw and Print

- Choose pens, surface
- Ensure alignment
- Print from Inkscape Dialog



INKSCAPE
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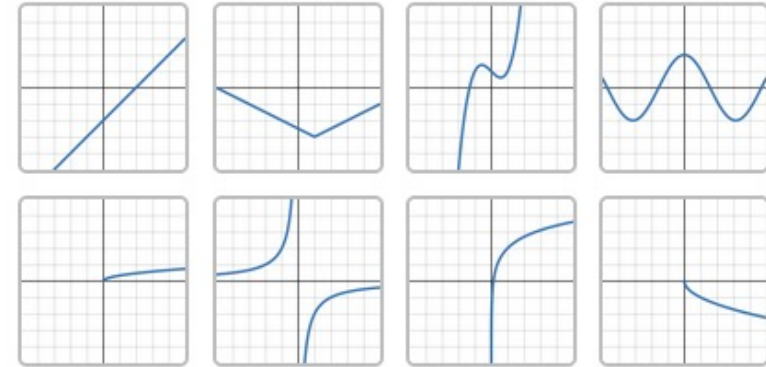


GETTING TO SUCCESS: CONTENT OBJECTIVES

Math Content:

- Develop function sense
 - Basic parent functions

I teach mathematics.

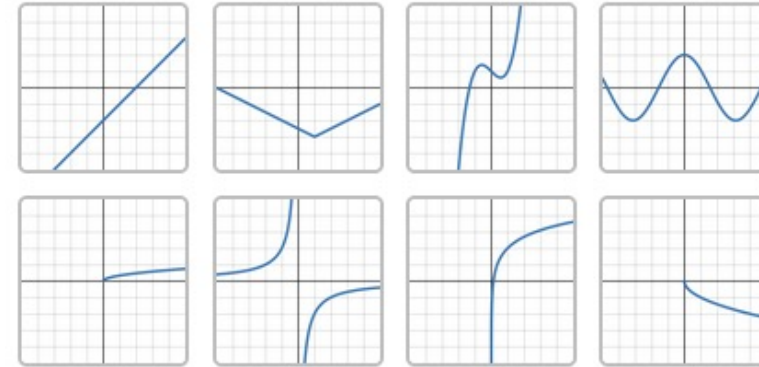


GETTING TO SUCCESS: CONTENT OBJECTIVES

Math Content:

- Develop function sense
 - Basic parent functions
- Transform, Combine functions
 - Translation, Dilation
 - Understand the graphs of $|f(x)|$, $f(x)+g(x)$, $f(x)*g(x)$, $f(g(x))$

I teach mathematics.

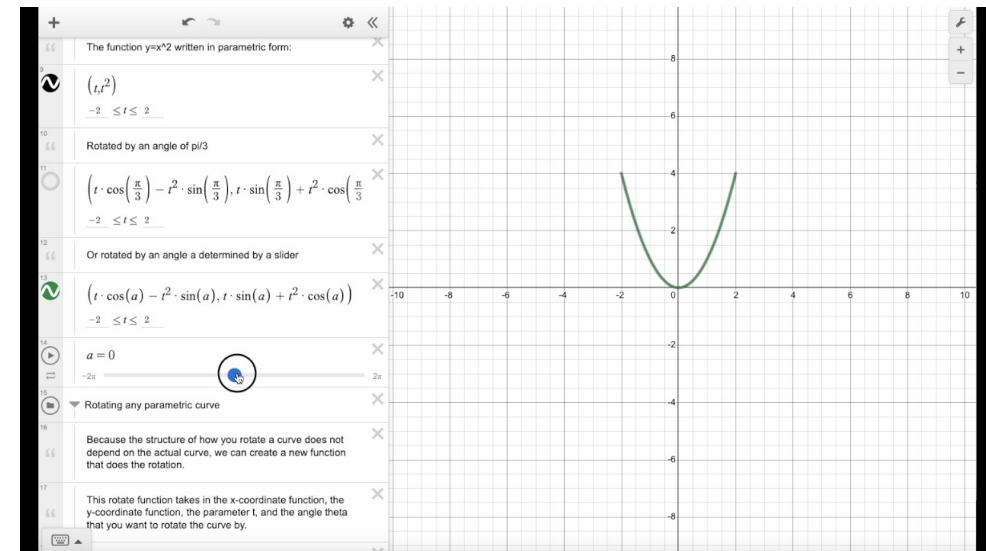
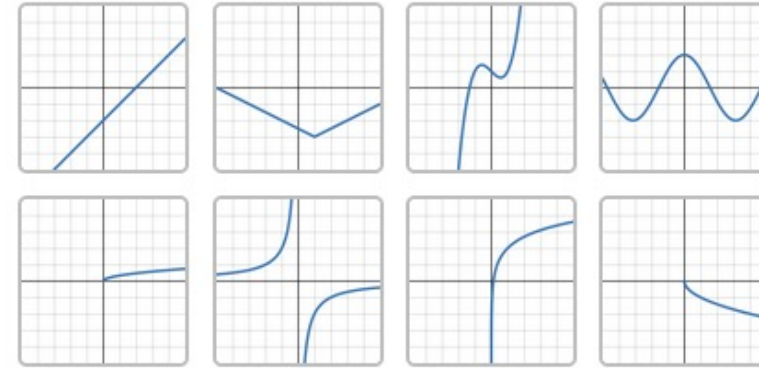


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 - Understand the graphs of $|f(x)|$, $f(x)+g(x)$, $f(x)*g(x)$, $f(g(x))$
- NEW: Polar and Parametric curves
 - Rotate the graph of $y=x^2$

I teach mathematics.



GETTING TO SUCCESS: CONTENT OBJECTIVES

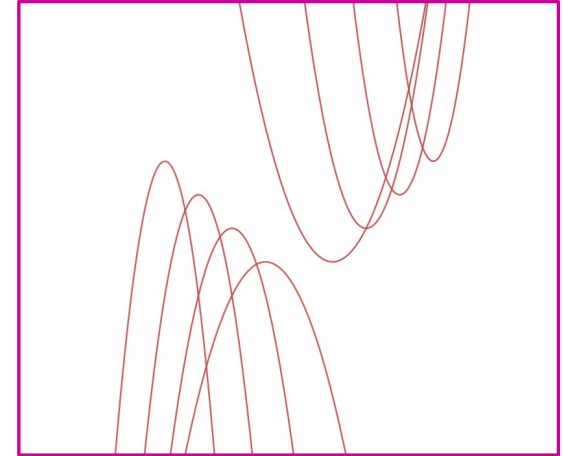
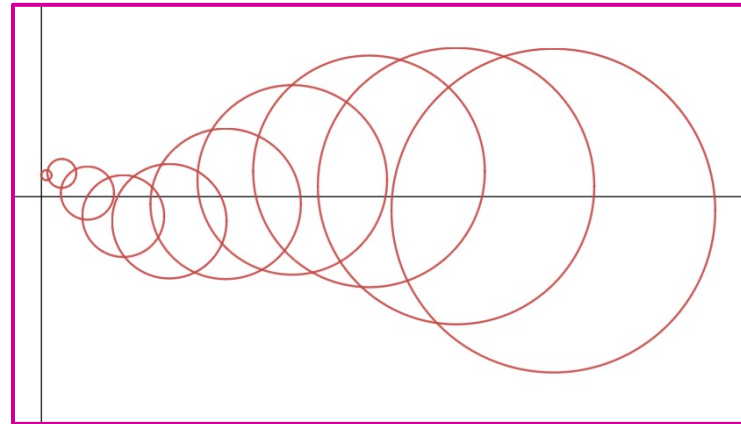
Math Content:

- Develop function sense
- Transform, Combine functions
- Polar and Parametric curves

Coding Content:

- Graphing interface
- Lists and Algorithmic techniques

I teach programming.



GETTING TO SUCCESS: CONTENT OBJECTIVES

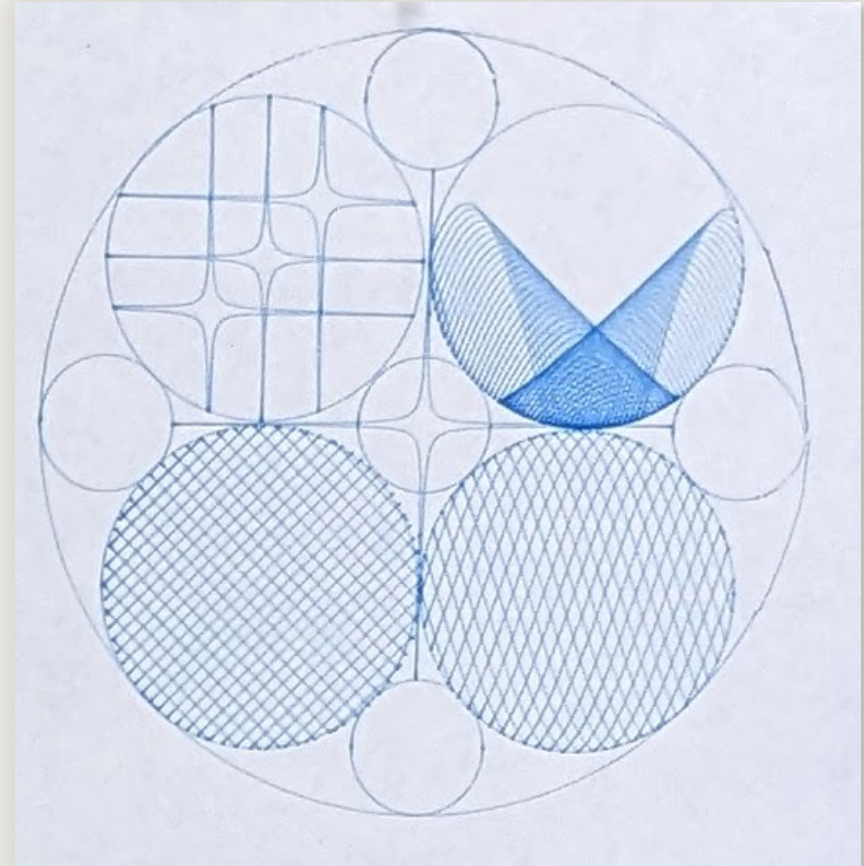
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Coding Content:

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- Restrictions of functions

I teach programming.



Richard Son Spr. 2023

GETTING TO SUCCESS: CONTENT OBJECTIVES

Math Content:

- Develop function sense
- Transform, Combine functions
- Polar and Parametric curves

I teach art.

Elements of Art

Principles of Design

Coding Content:

- Graphing interface
- Lists and Algorithmic techniques
- Restrictions of functions

Art Theory:

- Art and Design

https://www.getty.edu/education/teachers/building_lessons/formal_analysis.html

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Elements of Art

Principles of Design



Line



Shape



Space



Color

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I teach art.

Elements of Art



Line



Shape



Space



Color

Principles of Design



Movement



Pattern



Rhythm



Unity

https://www.getty.edu/education/teachers/building_lessons/formal_analysis.html

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From **Exploration**

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From **Exploration**
Through **Creativity**

GETTING TO SUCCESS: CONTENT OBJECTIVES

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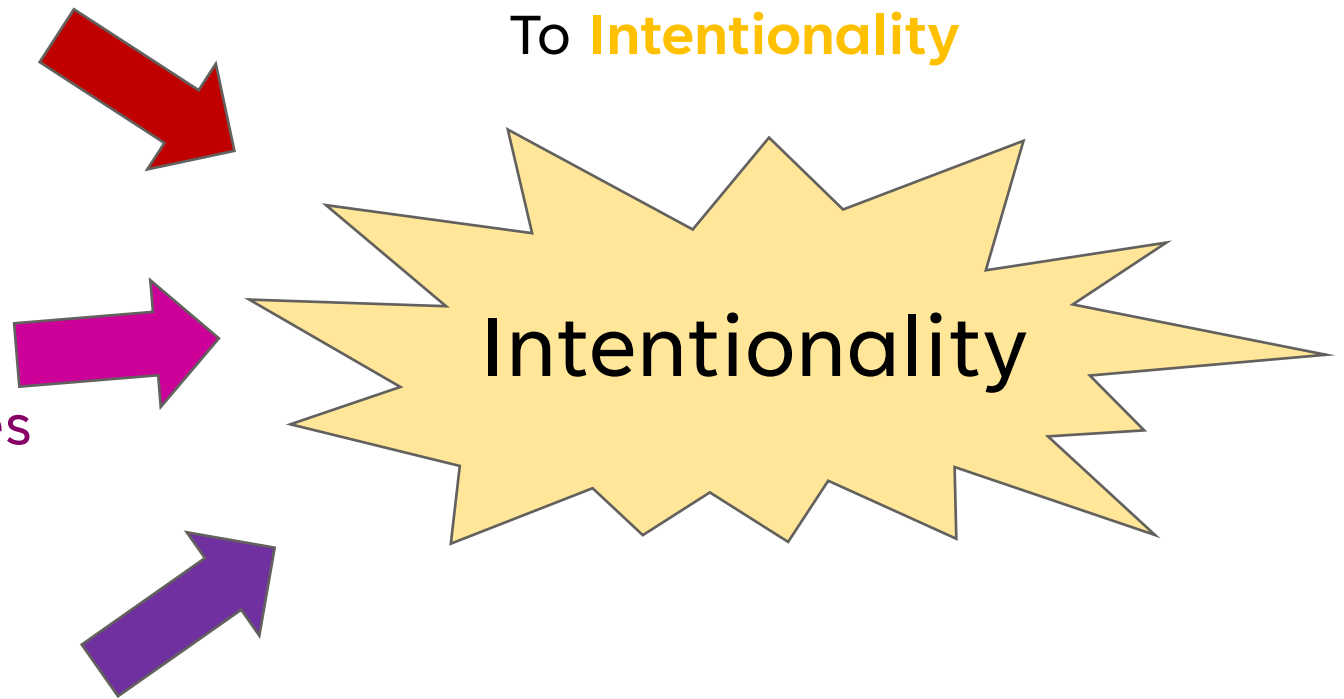
Coding Content:

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From **Exploration**
Through **Creativity**
To **Intentionality**



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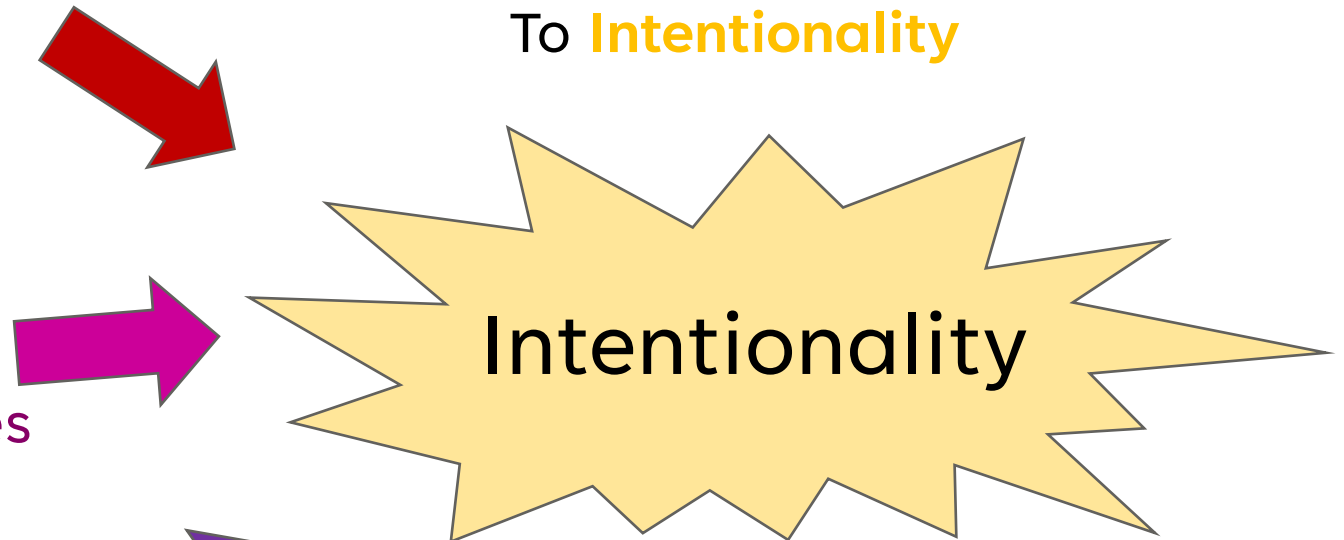
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Art Theory:

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From **Exploration**
Through **Creativity**
To **Intentionality**



Intentionality is the focus
of the written report.

I teach writing.

GETTING TO SUCCESS: SKILL OBJECTIVES



Content is not enough. Students need additional skills to succeed.

Technical Skills:

Design Thinking Skills:

Life skills:

GETTING TO SUCCESS: SKILL OBJECTIVES



Content is not enough. Students need additional skills to succeed.

Technical Skills:

- Software / Hardware Interface
- Debugging code

Design Thinking Skills:

Life skills:

GETTING TO SUCCESS: SKILL OBJECTIVES



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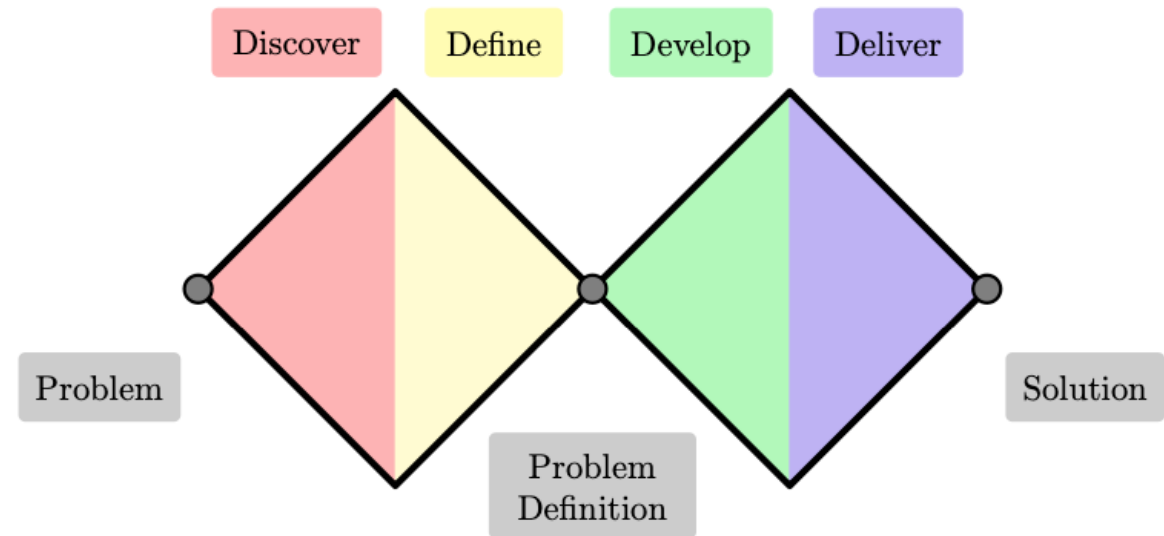
Technical Skills:

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Design Thinking Skills:

- Brainstorming / Culling
- Problem Definition
- Prototyping / Refining

Life skills:



Double Diamond Design model

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Technical Skills:

- Software / Hardware Interface
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Life skills:

- Thinking / Talking about what they are doing
- Getting help from classmates
- **Individual** progress in a supportive **Group** environment



PROJECT-BASED LEARNING

Deliverables: Students complete three projects and assemble a portfolio.

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Project 1. Create a sketch with transformations.

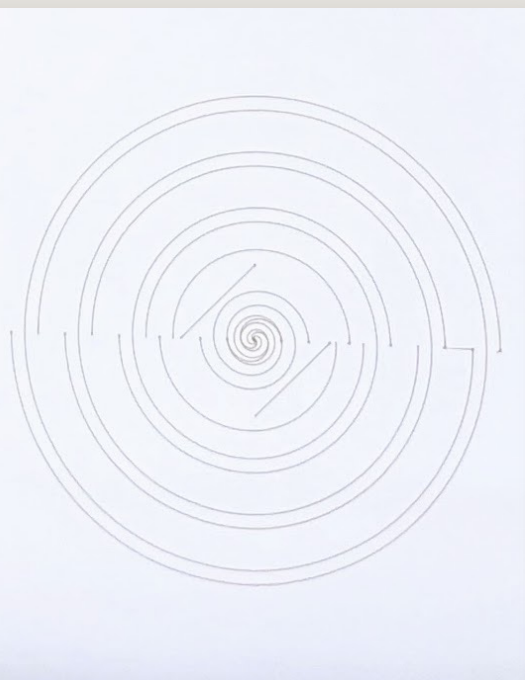
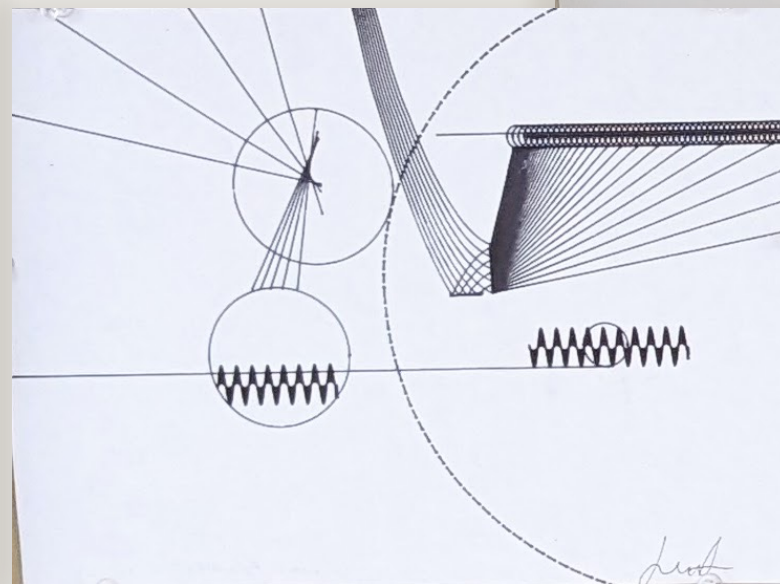
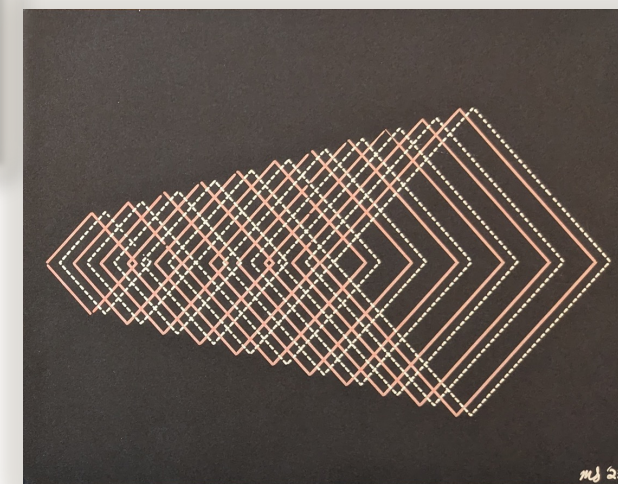
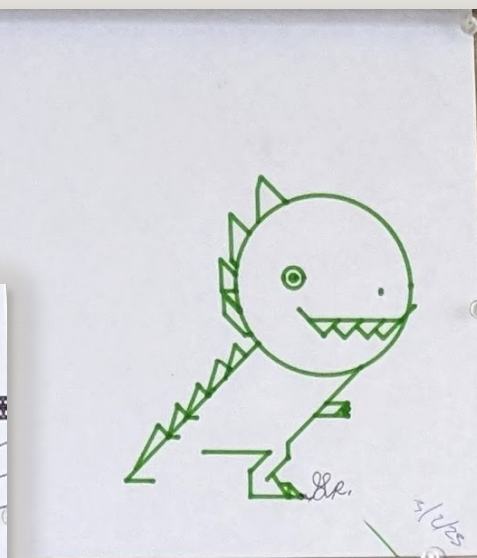
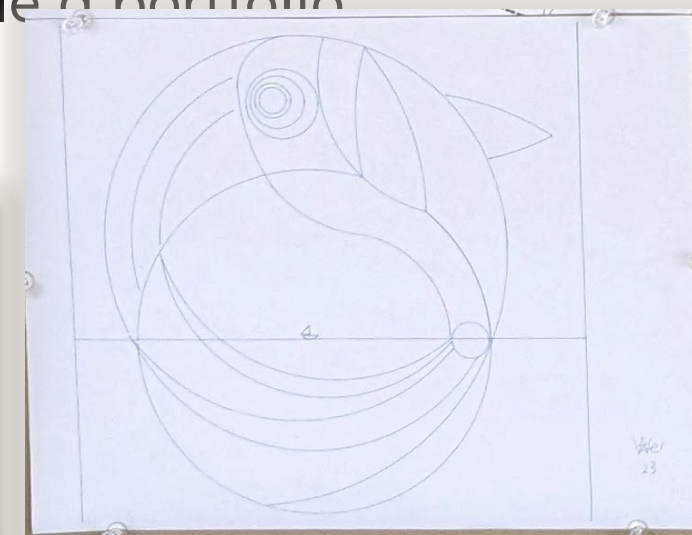
- Basic skills in Desmos
- Basic materials on the AxiDraw

PROJECT-BASED LEARNING

Deliverables: Students complete three projects and assemble a portfolio

Project 1. Create a sketch with transformations.

- Basic skills in Desmos
- Basic materials on the AxiDraw



PROJECT-BASED LEARNING

Deliverables: Students complete three projects and assemble a portfolio.

Project 1. Create a sketch with transformations.

Project 2. Push the AxiDraw further.

- Intermediate mathematics and skills in Desmos
- More advanced materials on the AxiDraw

PROJECT-BASED LEARNING

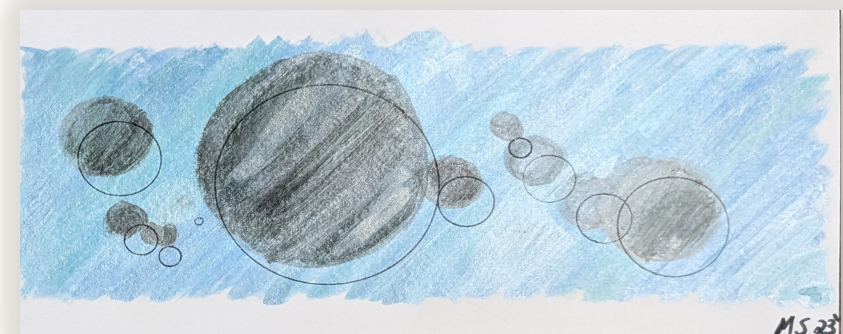
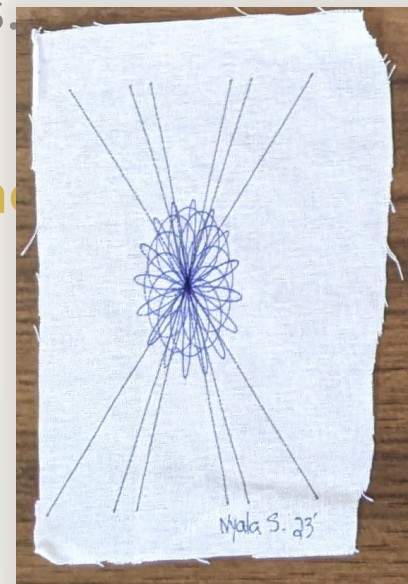
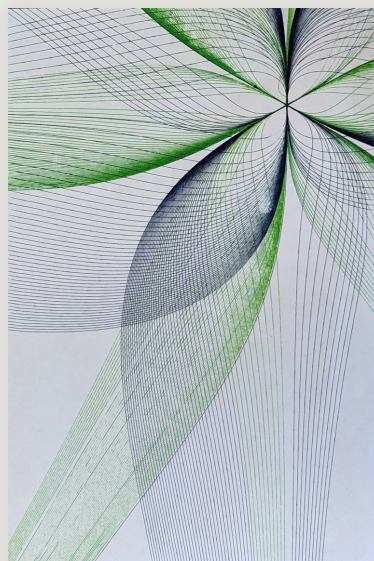
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→ Intermediate mathematics and skills in Desmos

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PROJECT-BASED LEARNING

Deliverables: Students complete three projects and assemble a portfolio.

Project 1. Create a sketch with transformations.

Project 2. Push the AxiDraw further.

Project 3. Explore possibilities in the Queens College Makerspace

→ Advanced mathematics

→ New machines

PROJECT-BASED LEARNING

Deliverables: Students complete three projects and assemble a portfolio.

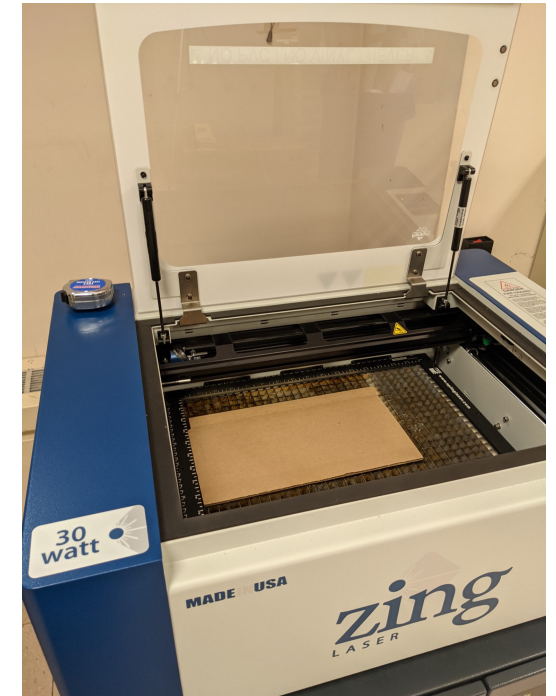
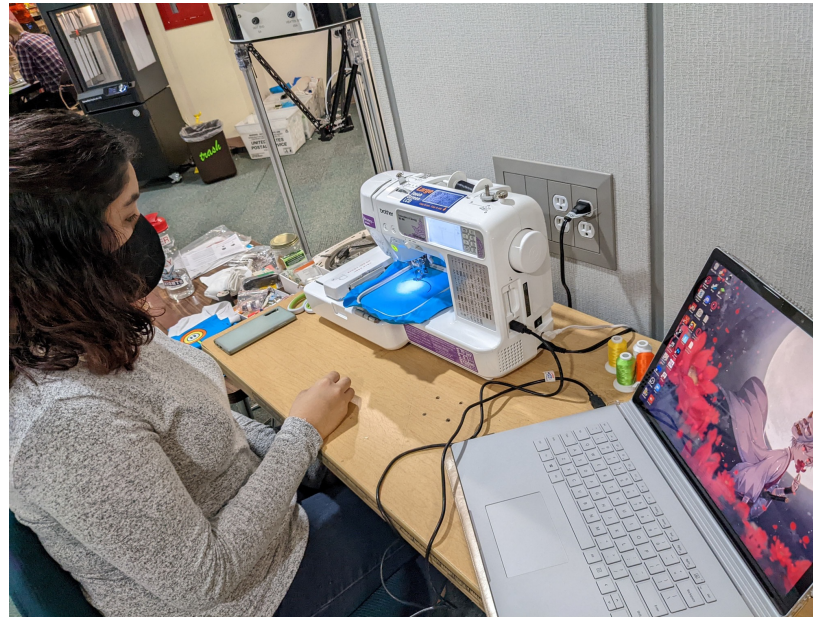
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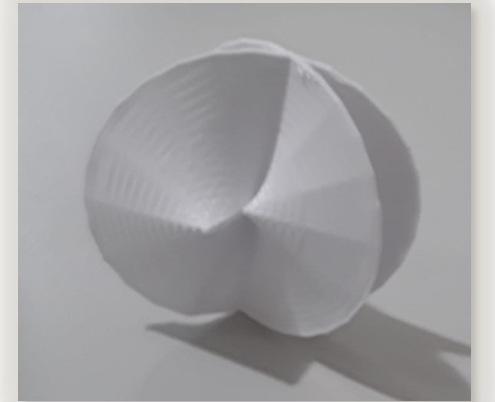
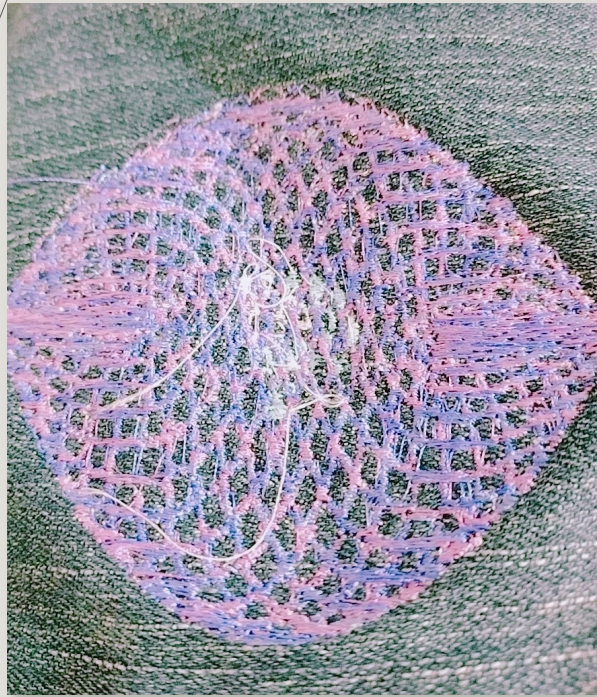
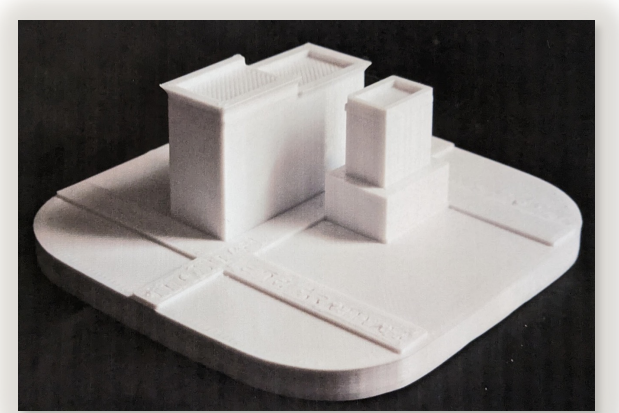
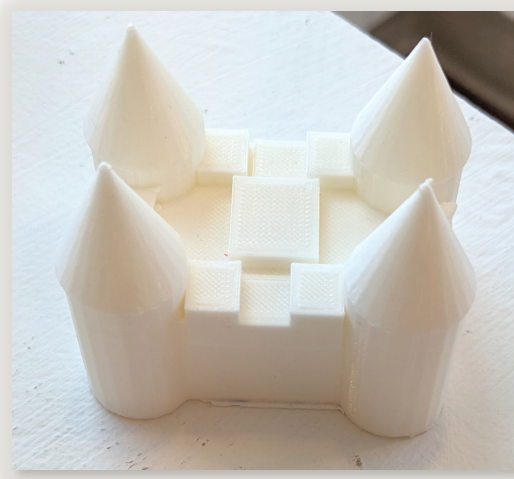
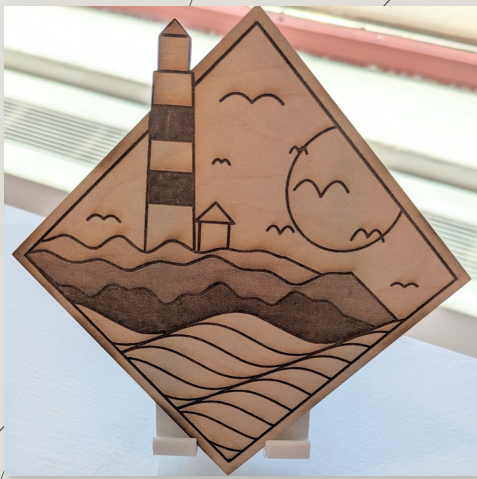
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September 28, 2023

GRADING PROJECTS

Convey Expectations through List of Standards; Score each one.

Engagement:

- Steady progress; attend class; turn in on time

Intentionality:

- Deliberate creation; matches artistic terms; honed over time

Desmos Notebook Content:

- Use techniques from class? Apply the mathematics we learned?

Desmos Notebook Style:

- Organization & Documentation of File

Process Discussion:

- How did your piece come to be / evolve? What hurdles or successes were there?

Technical Discussion:

- Explain the math, programming you used in technical language

Artistic Discussion:

- Explain the artistic qualities you aimed for in technical language

Writing style and format:

- Use good English, writing flows well, Correct Formatting

WHAT I LOVE ABOUT PROJECTS

For the Students

- Active Learning
- APPLY the content
- Authentic selves;
Harness the diversity!
- Accessible for students at
ALL LEVELS of understanding.

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For the Instructor

- Non-content-based
Learning Objectives
- A chance for me to learn:
 - Students approach things
differently than I would.
 - Each class is different.
 - I get to learn new skills.

BEST PRACTICES: BUILD SCAFFOLDING



BEST PRACTICES: BUILD SCAFFOLDING

Provide a transparent timeline.

- **Gain Expertise** by Thursday, February 9.
- **Explore Possibilities** by Thursday, February 16.
- **Peer Review** on Tuesday, February 28.
- **Submit Final Project** on Thursday, March 2.



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Include In-class work time

Peer Review: (Early deadline; what works / what doesn't)



BEST PRACTICES: BUILD SCAFFOLDING

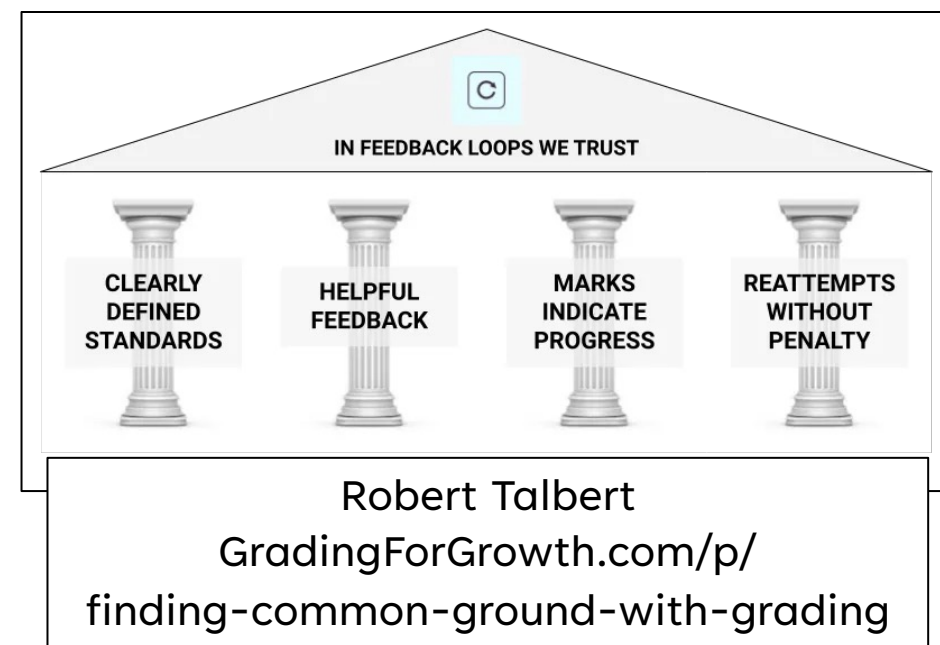
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- **Submit Final Project** on Thursday, March 2.

Include In-class work time

Peer Review: (Early deadline; what works / what doesn't)

Give (prompt) **Detailed Feedback** and an **Opportunity to Revise** to improve scores.



BEST PRACTICES: BUILD SCAFFOLDING

Provide a transparent timeline.

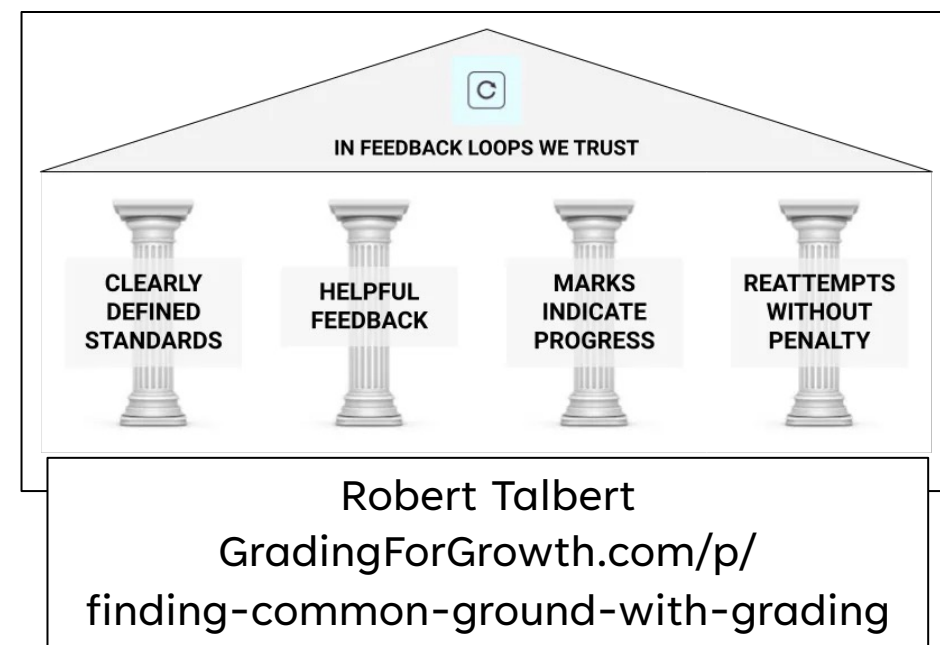
- **Gain Expertise** by Thursday, February 9.
- **Explore Possibilities** by Thursday, February 16.
- **Peer Review** on Tuesday, February 28.
- **Submit Final Project** on Thursday, March 2.

Include In-class work time

Peer Review: (Early deadline; what works / what doesn't)

Give (prompt) **Detailed Feedback** and an **Opportunity to Revise** to improve scores.

Celebrate their progress



REFLECTION AND CELEBRATION

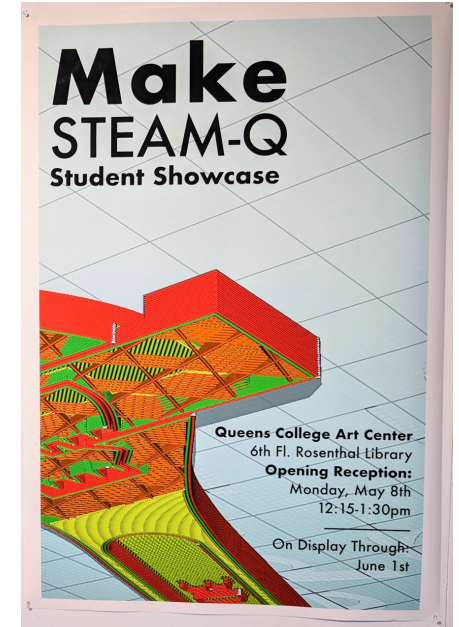
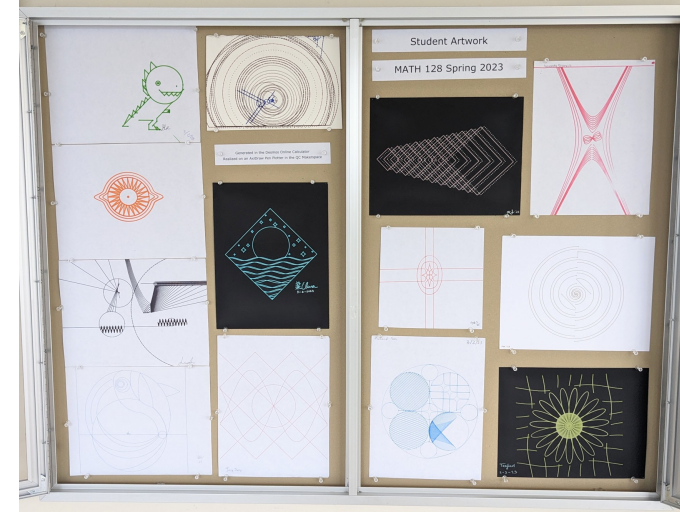
Final Project: Portfolio and Reflection

- Accumulate Drafts
- Present their work – Tell a story!
- Growth as a **mathematician.**
- Growth as a **maker.**
- Growth as an **artist.**
- Give students agency over their voice.

REFLECTION AND CELEBRATION

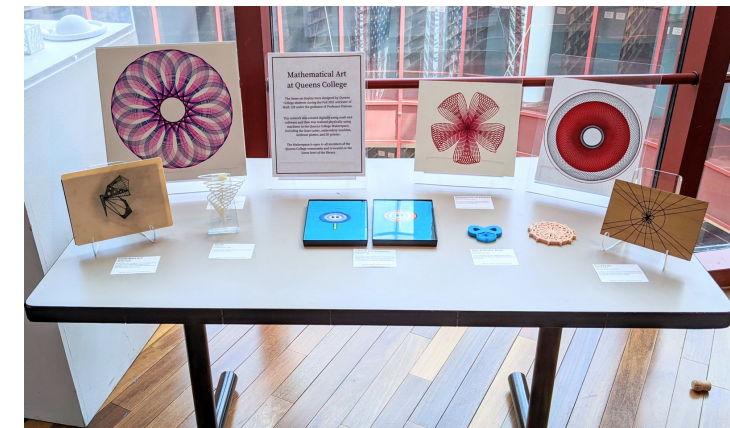
Final Project: Portfolio and Reflection

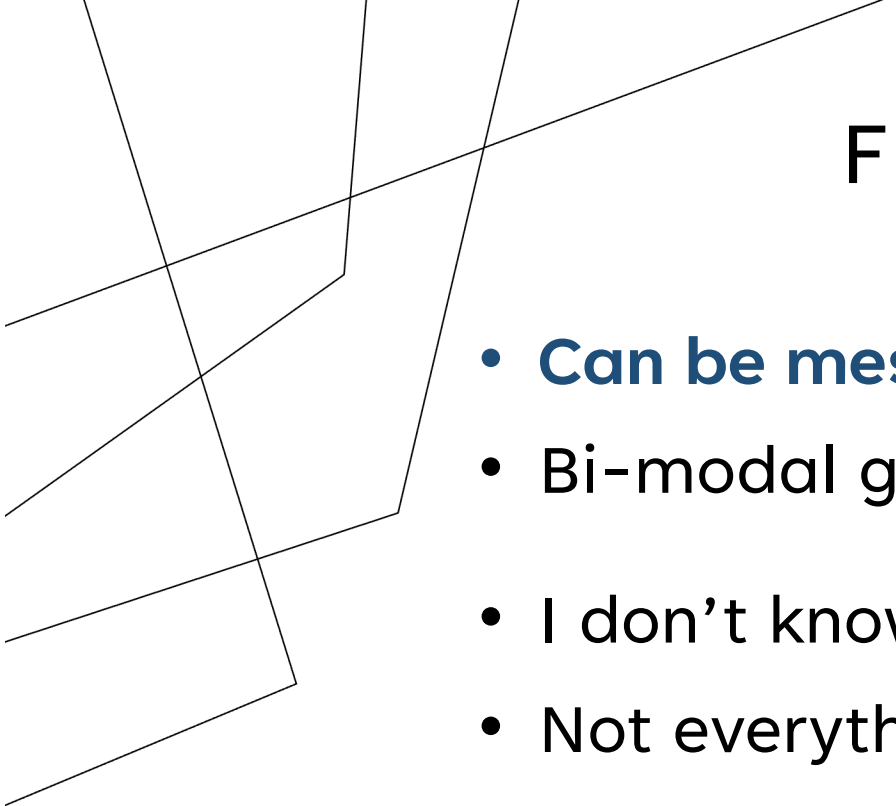
- Accumulate Drafts
- Present their work - Tell a story!
- Growth as a **mathematician**.
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- Growth as an **artist**.
- Give students agency over their voice.



Celebration: Art Exhibition

- Feel like an artist!
- Gallery on campus
- Opening Reception
- Share with friends!





FINAL THOUGHTS

- **Can be messy** – be flexible (rules; students' lives)
- Bi-modal grade distributions
- I don't know everything.
- Not everything that works for me will work for you.
- It won't be perfect the first time and **that's OK**.



THANK YOU!
I AM: **CHRISTOPHER HANUSA**

Website qc.edu/~chanusa

- > Courses > Course Archive **(syllabi)**
- > Research > Talks **(slides)**

Email chanusa@qc.cuny.edu

Social @mathzorro / @hanusadesign

Jewelry hanusadesign.com



THANK YOU!

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Jewelry hanusadesign.com

Illustrating Mathematics Discord
discord.gg/jCvndAPfh

Unofficial Desmos Discord
discord.gg/vCBupKs9sB

Alternative Grading Slack
thegradingconference.com/resources/

Grading For Growth(.com)
Robert Talbert and David Clark